



PREDICTION OF THE TOTAL COST OF STUDYING ABROAD

GROUP 05

PREPARED FOR:

DS 3003 / ST 3082
Data Analysis Project 02
Advanced Analysis

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Abstract

The cost of studying abroad is a key yet uncertain factor affecting students' decisions. This project developed a machine learning framework to predict the total cost of international education using academic and economic variables. A dataset of 907 records with 12 features (country, city, university, program, academic level, duration, living cost index, tuition, rent, visa fees, insurance, and exchange rates) was analyzed using a full preprocessing pipeline with cleaning, feature engineering, encoding, and scaling. The total cost was calculated by combining tuition, living expenses, rent, visa, insurance, and duration. Several models were tested, and after optimization and SHAP-based feature selection, the CatBoost model achieved strong performance ($R^2 \approx 0.95$). The project produced the EduCost Card Builder, an interactive tool to help students estimate personalized study-abroad expenses.

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1. Introduction

“Learning beyond borders is a gateway to growth and endless possibilities.”

International education has become an increasingly popular pathway for students seeking global exposure, high-quality academic opportunities, and improved career prospects. However, planning to study abroad requires substantial financial preparation, and many students struggle to estimate the true overall cost of their education beyond tuition fees. Expenses such as living costs, accommodation, insurance, visa fees, and currency exchange fluctuations significantly influence the total investment required. This study aims to address this challenge by applying exploratory data analysis and advanced machine learning techniques to understand global cost patterns and develop a reliable model for predicting the total cost of an international education journey.

2. Description of the question

Studying abroad provides valuable academic and career opportunities, yet estimating the true financial commitment is challenging for many students. In addition to tuition, costs such as living expenses, rent, visa fees, insurance, and exchange rate fluctuations greatly affect the total expense, and global surveys show that *only about 55% of students understand the real financial requirements*. This study addresses *how to accurately predict the total cost of international education based on students' choices by developing a machine learning model using key factors such as country, academic level, program, duration, exchange rate, and living cost index, supported by exploratory data analysis and feature selection to identify the most influential predictors*.

3. Description of the Dataset

Dataset name: Cost of International Education

link: <https://www.kaggle.com/datasets/adilshamim8/cost-of-international-education>

The Cost of International Education Dataset obtained from Kaggle contains 907 observations with 12 variables of which 7 are numeric and 5 are categorical. Description of each variable can be found in the table below:

Table 1: Description of Variables

Variable Name	Type	Description
Country	Qualitative (Nominal)	ISO country name where the university is located. The dataset includes universities from 71 different countries .
City	Qualitative (Nominal)	The city in which the institution is located. The dataset includes 56 unique cities .
University	Qualitative (Nominal)	Official name of the higher-education institution. The dataset includes 622 unique universities .

Program	Qualitative (Nominal)	Specific course(degree) name.The dataset includes 92 distinct programs
Level	Qualitative (Ordinal)	Degree level of the program (Bachelor,Master, PhD)
Duration Years	Quantitative (Discrete)	Length of the program in years
Tuition USD	Quantitative (Continuous)	Total program tuition cost, converted into U.S. dollars for ease of comparison.
Living Cost Index	Quantitative (Continuous)	A normalized index of daily living costs (food, transport, utilities), benchmarked to New York City = 100 , based on global cost-of-living data.
Rent USD	Quantitative (Continuous)	Average monthly student accommodation rent in U.S. dollars.
Visa Fee USD	Quantitative (Continuous)	One-time visa application fee payable by international students, in U.S. dollars.
Insurance USD	Quantitative (Continuous)	Annual health or student insurance cost in U.S. dollars, as required by many host countries.
Exchange Rate	Quantitative (Continuous)	Local currency units per U.S. dollar at the time of data collection.

4. Before EDA

4.1. Data Preprocessing

Before conducting the EDA, the dataset was carefully cleaned and assessed for quality. No missing values or duplicate records were identified. All variables were verified for appropriate data types, and no modifications were necessary. The data was also screened for unrealistic or inconsistent values, with none detected. Summary statistics were reviewed to confirm overall accuracy and consistency.

4.2. Feature Engineering

- The target variable, Total Cost USD, was created which represents the total cost of studying abroad over the full program duration by combining major cost components.

$$\text{Total_Cost_USD} = \text{Tuition_USD} + (\text{Living_Cost_Index}/100 \times 12000 \times \text{Duration_Years}) + (\text{Rent_USD} \times 12 \times \text{Duration_Years}) + \text{Visa_Fee_USD} + (\text{Insurance_USD} \times \text{Duration_Years})$$

The Cost of Living Index (COLI) is benchmarked to New York City (100). To translate this into an absolute monetary value, a baseline annual living cost of USD 12,000 was assumed, based on 2025 estimates of essential non-accommodation expenses in New York City.

Particulars	Cost (Monthly)
Off-Campus Accommodation	\$1,050
On-Campus Accommodation	\$1,252
Transportation	\$128
Food Cost	\$638
Utility Cost	\$116
Entertainment	\$65
Groceries	\$116
Clothes	\$42
Health Insurance	\$80
Total Cost of Living with Off-Campus Accommodation	\$2,027
Total Cost of Living with On-Campus Accommodation	\$2,229
Annual Cost of Living	\$24,324

Figure 1: 2025 estimates of essential non- accommodation expenses in New York City

Monthly Total = Transportation + Food + Utilities + Groceries + Clothes
 $= 128 + 638 + 116 + 116 + 42$
 $= 1,040 \text{ USD/month}$

Annual Total = Monthly Total $\times 12 = 1,040 \times 12 = 12,480 \text{ USD/year}$
Rounded Approximation $\approx 12,000 \text{ USD/year}$

4.3. Train Test Split of the Dataset

- The dataset was divided into training(70%) and testing(30%) datasets

The training dataset has 634 rows and 13 columns.
The testing dataset has 273 rows and 13 columns.

Figure 2: Train Test Split

4.4. Results of FAMD

Factor Analysis of Mixed Data (FAMD) was applied for dimensionality reduction, but capturing 90% of the variance required at least 7 factors. As this provides minimal reduction from the original dataset, FAMD was deemed ineffective for this model.

4.4.1. Clustering

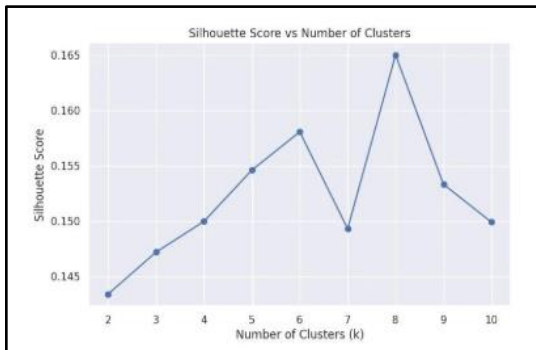


Figure 4: Plot of silhouette scores vs. Number of Clusters

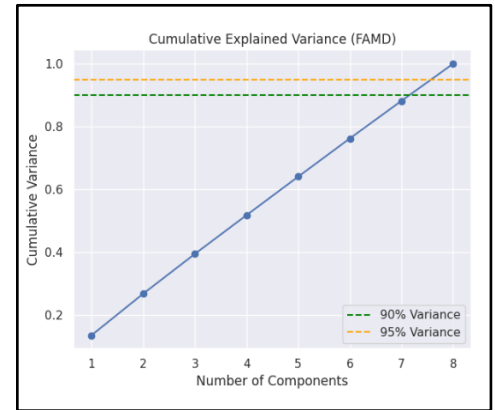


Figure 3: Scree Plot of Cumulative Variance (FAMD)

Evaluating K-Means clustering for 2–10 clusters showed that the most distinct cluster structure occurs at $k = 8$, based on the silhouette scores.

Although this was the highest score within the tested range ($k = 2$ to $k = 8$), all silhouette values were below 0.2. This indicates substantial overlap between clusters, suggesting that the data does not naturally form well-separated groups and no significant clustering structure is present.

4.5. Results of Multicollinearity Analysis

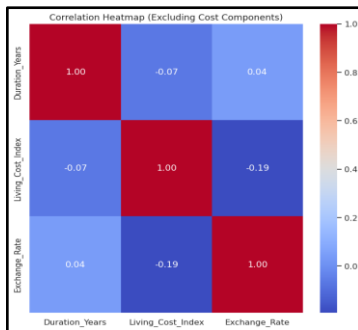


Figure 6: Correlation Heatmap of Duration, Living Cost Index, and Exchange Rate

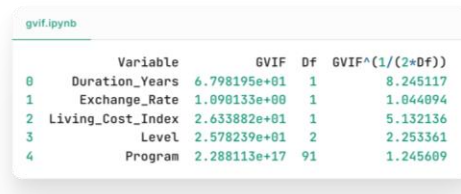


Figure 7: GVIF Analysis for Multicollinearity Detection

Analysis of the correlation heatmap and GVIF results shows **no significant multicollinearity** among the predictors.

5. Important results of the Descriptive Analysis

Students usually first think about their level of study. This graph shows that Bachelor's programs have higher median tuition than Master's or PhD programs. But when looking at the total cost; including living expenses, Bachelor's and PhD programs are almost equal on average. PhD programs, however, can become much more expensive due to their longer duration and accumulating living costs.

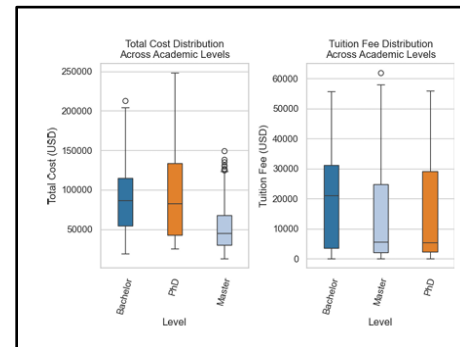


Figure 8: Boxplots of Total Cost and Tuition Fee vs Academic Levels

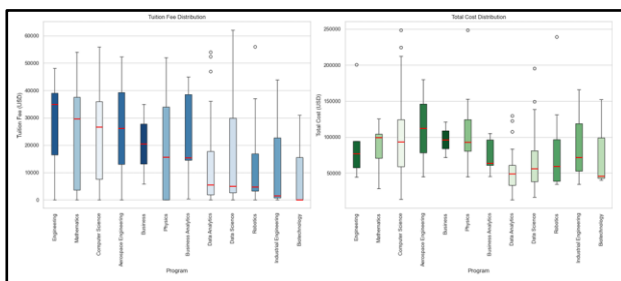


Figure 9: Boxplots of Tuition Fee and Total Cost vs Academic Program

Our analysis shows that most STEM fields, like Engineering, Mathematics, and Computer Science, have higher median tuition and wider price ranges than other subjects. This means that while the field of study strongly affects tuition, it is less predictive of the total cost, though it still plays a role.

Analysis across countries shows that rent and living expenses are usually the biggest cost for international students. While tuition matters, in many places the total cost of housing and daily expenses can exceed the tuition itself. The country a student studies in strongly influences the total cost due to large regional differences.

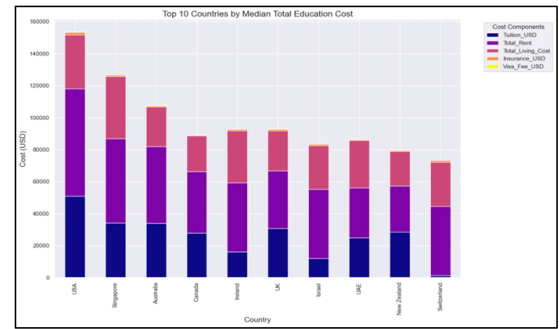


Figure 10: Stacked Bar Chart of Total Education Cost Components vs Top 10 Countries

5.1. EDA results for Advanced Modeling

Based on these EDA insights, we incorporate **Country, Program, and Academic Level** as key predictors in our advanced cost prediction model.

6. Advanced Modeling procedure

```

categorical_transformer = Pipeline(steps=[
    ('onehot', OneHotEncoder(handle_unknown='ignore'))
])

numerical_transformer = Pipeline(steps=[
    ('scaler', StandardScaler())
])

preprocessor = ColumnTransformer(
    transformers=[
        ('cat', categorical_transformer, categorical_features),
        ('num', numerical_transformer, numerical_features)
    ])

# Define cross-validation strategy (5-fold CV for regression)
cv = KFold(n_splits=5, shuffle=True, random_state=42)
mae_scorer = make_scorer(mean_absolute_error, greater_is_better=False)

# Define models and their hyperparameter grids for tuning
models_config = {
    "Linear Regression": {
        "model": LinearRegression(),
        "params": None, # No hyperparameters to tune
    },
    "Ridge": {
        "model": Ridge(random_state=42),
        "params": {'model__alpha': [0.001, 0.01, 0.1, 1.0, 10.0, 100.0]},
    },
    "Lasso": {
        "model": Lasso(random_state=42, max_iter=2000),
        "params": {'model__alpha': [0.0001, 0.001, 0.01, 0.1, 1.0, 10.0]},
    },
    "Elastic Net": {

```

Figure 11: Pipeline for Multi-Model Regression and Hyperparameter Tuning

We used a structured modeling pipeline to ensure **consistent preprocessing and optimization** across all models. **Categorical features were one-hot encoded and numerical features were standardized.** Model performance was evaluated using **5-fold cross-validation** for reliable results. Each model underwent **hyperparameter tuning** (e.g., tree depth, number of estimators, learning rate, regularization), ensuring fair comparison and robust prediction of international education costs.

7. Initial modeling

In the Initial Modeling phase, the main goal was to establish a baseline for predicting total cost and identify the best way to use key variables. Initially, the model included the Living Cost Index directly in the target variable (Total_Cost_USD), but later it was used as an independent predictor instead.

- **Living Cost Index in the Target Variable(Total_Cost_USD)**

At the beginning of the study, the Living Cost Index was directly included into the target variable(Total_Cost_USD), through a predefined cost function.

$$\text{Total_Cost_USD} = \text{Tuition_USD} + (\text{Living_Cost_Index}/100 \times 12000 \times \text{Duration_Years}) + (\text{Rent_USD} \times 12 \times \text{Duration_Years}) + \text{Visa_Fee_USD} + (\text{Insurance_USD} \times \text{Duration_Years})$$

- **Living Cost Index as a Predictor**

In the next stage, the Living Cost Index was removed from the target variable and it was treated as an independent predictor variable in the modeling process.

$$\text{Total_Cost_USD} = \text{Tuition_USD} + (\text{Rent_USD} \times 12 \times \text{Duration_Years}) + \text{Visa_Fee_USD} + (\text{Insurance_USD} \times \text{Duration_Years})$$

In the final stage of this study, a component based modeling approach was applied to improve the accuracy and interpretability of total cost estimation, rather than using a single model to predict total cost. Individual cost components were modeled separately using multivariate regression techniques.

7.1. Modeling (taking Living Cost as a part of response)

Model Comparison (Train vs Test):			
	Train_RMSE	Train_R2	Test_RMSE \
Linear Regression	6579.519448	0.916970	6554.522446
Ridge	6582.773550	0.916888	6517.469326
Lasso	6582.685614	0.916890	6495.144864
Elastic Net	6590.446258	0.916694	6499.736213
Decision Tree	5332.424827	0.945462	6725.400308
Random Forest	5564.744796	0.940607	6477.922120
Gradient Boosting (Boosted Trees)	5486.488907	0.942265	6579.804705

	Test_R2
Linear Regression	0.925132
Ridge	0.925976
Lasso	0.926482
Elastic Net	0.926378
Decision Tree	0.921178
Random Forest	0.926872
Gradient Boosting (Boosted Trees)	0.924553

Best model based on Test RMSE: Random Forest
Test RMSE: \$6477.92
Test R2: 0.9269

The **Random Forest** is the best performing model with lowest Test RMSE (\$6477.92) and the highest Test R -Squared (0.9269). However, overfitting was evident, as the training error (\$5,564) was much lower than the test error.

Figure 12: Model Comparison and Selection
(Living Cost Included in Response)

7.2. Modeling (taking Living Cost as a predictor)

Model Comparison (Train vs Test):			
	Train_RMSE	Train_R2	Test_RMSE \
Linear Regression	5287.420785	0.940851	5435.088222
Ridge	5290.688100	0.940778	5397.086303
Lasso	5292.045483	0.940747	5392.532789
Elastic Net	5310.684744	0.940329	5365.781957
Decision Tree	3120.850096	0.979393	6882.692204
Random Forest	2650.131263	0.985141	5084.841482
Gradient Boosting (Boosted Trees)	1835.407123	0.992873	4779.063461

	Test_R2
Linear Regression	0.944080
Ridge	0.944859
Lasso	0.944952
Elastic Net	0.945497
Decision Tree	0.910325
Random Forest	0.951055
Gradient Boosting (Boosted Trees)	0.956765

Best model based on Test RMSE: Gradient Boosting (Boosted Trees)
Test RMSE: \$4779.06
Test R2: 0.9568

Figure 13: Model Comparison and Selection
(Living Cost as Predictor)

Here, the Living Cost Index was treated as an independent predictor. This improved accuracy, with **Gradient Boosting** performing best (Test R² = 0.9568, Test RMSE = \$4,779), though some overfitting still remained.

The key insight is that treating the Living Cost Index as a feature is better for modeling.

8. Modeling (Outlier Handling)

8.1. Univariate Outliers

	Variable	Number_of_Outliers	Percentage_of_Outliers (%)
0	Duration_Years	0	0.00
1	Tuition_USD	0	0.00
2	Living_Cost_Index	20	3.15
3	Rent_USD	3	0.47
4	Visa_Fee_USD	93	14.67
5	Insurance_USD	78	12.30
6	Exchange_Rate	169	26.66
7	Total_Cost_USD	23	3.63

Figure 14: Univariate Outliers

The data shows a wide range of stability across different costs, with some variables being much more consistent than others. Both `Duration_Years` and `Tuition_USD` recorded zero outliers, meaning these values stayed very steady throughout the dataset. On the other hand, the `Exchange_Rate` was the most unpredictable factor, with over **26%** of its data points flagged as outliers. We also saw high variation in `Visa_Fee_USD` and `Insurance_USD`, which had outlier rates of **14.67%** and **12.30%** respectively. Meanwhile, the `Living_Cost_Index`, `Total_Cost_USD`, and `Rent_USD` all remained relatively stable, with only a small handful of unusual values.

8.2. Modeling without outliers

Once we removed the individual outliers, the **R² value dropped significantly**, which proves that these outliers actually carry important information for the model. Even though they are unusual data points, they seem to be a key part of the overall pattern rather than just "noise" to be deleted.

Model comparison (Train vs Test):			
	Train_RMSE	Train_R2	Test_RMSE \
Linear Regression	5458.660604	0.889278	6226.758401
Ridge	5461.225324	0.889174	6191.145129
Lasso	5517.397044	0.886883	6060.294567
Elastic Net	5482.114020	0.888325	6130.524390
Decision Tree	1765.438831	0.988418	5402.091099
Random Forest	2487.430586	0.977009	4942.484384
Gradient Boosting (Boosted Trees)	1906.993166	0.986487	4884.692108
	Test_R2		
Linear Regression	0.862539		
Ridge	0.864107		
Lasso	0.869790		
Elastic Net	0.866755		
Decision Tree	0.896538		
Random Forest	0.913394		
Gradient Boosting (Boosted Trees)	0.915408		
Best model: Gradient Boosting (Boosted Trees)			
Test RMSE: \$4884.69			
Test R2: 0.9154			

Figure 15: Model Comparison and Selection (Without Outliers)

The reason these outliers are so important is that they aren't just "errors" or "mistakes" in the data; they represent **real-world extremes**. In our case, things like a sudden spike in an **Exchange_Rate** or a very high **Visa_Fee** are genuine situations that the model needs to understand.

By removing them, we are essentially "smoothing out" the data and hiding the reality of how expensive or volatile certain countries can be. By keeping them, the model learns to account for these high-cost scenarios, which makes its predictions much more accurate for the full range of actual experiences.

8.3. Multivariate Outliers

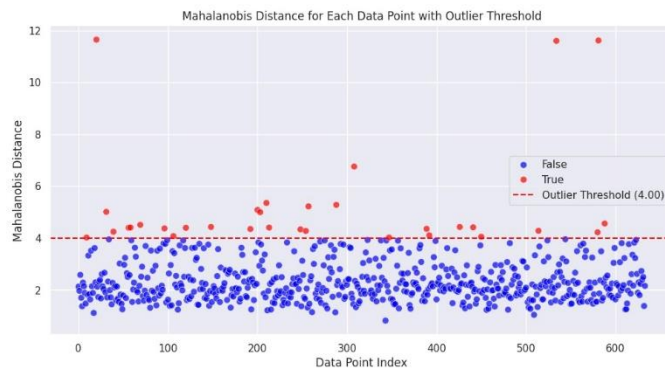


Figure 16: Multivariate Outliers

To detect outliers beyond individual variables, we applied **Mahalanobis Distance**, a multivariate method that considers all factors together. This identified **32 out of 907 points ($\approx 5.05\%$)** that exceeded the threshold of 4.002, indicating unusual combinations of tuition, exchange rates, and living costs. Such outliers can **skew model accuracy** if ignored, potentially biasing predictions toward rare extreme cases.

However, since they often reflect **real-world edge cases**, careful consideration is needed to decide whether to **transform or remove them**, ensuring reliable final predictions.

In our case, we decided to keep these outliers because our goal is to predict the total cost of studying abroad based on students' real-world choices. Some students intentionally choose unusually expensive destinations, premium universities, or countries with extreme living costs. Removing these cases would make the model less realistic and reduce its ability to provide accurate predictions for students with diverse study plans. Keeping them helps the model learn the full range of possible cost scenarios and makes the final predictions more practical and representative.

9. Feature selection using SHAP

We decided to try **SHAP (SHapley Additive exPlanations)** to see what kind of results we would get and identify which features are most significant. This gave us a clear look at how much each variable actually changed the final prediction. We found that while **Country** and **Exchange rate** were the main drivers, the **Living cost index** also had a significant impact on the outcome.

Model Comparison (Train vs Test):			
	Train_RMSE	Train_R2	Test_RMSE \
Linear Regression	2283.722784	0.988966	6350.194711
Ridge	3056.569743	0.980234	5296.557100
Lasso	3692.731752	0.971149	4755.846119
Elastic Net	2834.666820	0.982999	5238.946653
Decision Tree	2152.995189	0.990193	4610.959238
Random Forest	1639.053989	0.994316	3833.753074
Gradient Boosting (Boosted Trees)	942.288022	0.998121	3910.099625
	Test_R2		
Linear Regression	0.923664		
Ridge	0.946894		
Lasso	0.957184		
Elastic Net	0.948043		
Decision Tree	0.959753		
Random Forest	0.972177		
Gradient Boosting (Boosted Trees)	0.971058		
Best model based on Test RMSE: Random Forest			
Test RMSE: \$3833.75			
Test R2: 0.9722			

Figure 18: Model Comparison and Selection (With all variables)

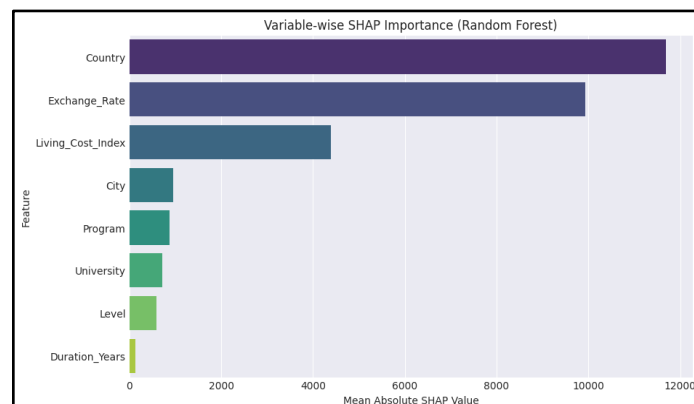


Figure 17: SHAP Feature Importance Ranking

10. Modeling after SHAP

Model Comparison (Train vs Test):			
	Train_RMSE	Train_R2	Test_RMSE \
CatBoost	3548.470916	0.973359	5226.860902
Random Forest	3629.403911	0.972130	5304.036662
Gradient Boosting (Boosted Trees)	3797.852847	0.969483	5309.115065
Ridge	5534.237034	0.935200	5386.387913
Lasso	5535.432553	0.935172	5390.646466
Elastic Net	5534.208528	0.935200	5391.484494
Decision Tree	4010.960086	0.965962	5672.468658
Linear Regression	5520.505474	0.935521	28652.532217

	Test_R2
CatBoost	0.948283
Random Forest	0.946744
Gradient Boosting (Boosted Trees)	0.946642
Ridge	0.945078
Lasso	0.944991
Elastic Net	0.944974
Decision Tree	0.939089
Linear Regression	-0.554102

Best model based on Test RMSE: CatBoost
Test RMSE: \$5226.86
Test R2: 0.9483
Best model saved as 'model.pkl'

Figure 19: Final Model Selection using Key Predictors

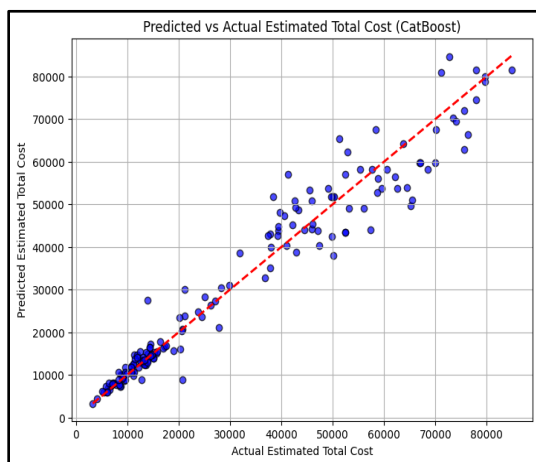


Figure 20: Scatter Plot of Predicted vs. Actual Values (CatBoost regressor)

Building on our SHAP analysis, we narrowed our focus to the most impactful features: **Country, Living Cost Index, and Exchange Rate**. By removing the less influential variables and concentrating on these core drivers, we were able to significantly refine the model's predictive power. This streamlined approach, combined with the **CatBoost** algorithm, ultimately achieved an impressive R^2 score of 0.95, indicating that these three factors alone explain **95%** of the variance in our data.

The scatter plot comparing predicted vs. actual total cost using the CatBoost model shows strong overall performance. Most data points cluster closely around the red diagonal line, indicating that the model's predictions are generally accurate and aligned with real values. As the actual cost increases, the spread of points becomes wider, suggesting slightly higher prediction error for more expensive study plans; this is a common pattern in regression because very high-cost cases appear less frequently in the training data. Importantly, there is no clear bias in the model, as the points are distributed fairly evenly above and below the line.

11. Multivariate modeling

The comparison of our **multivariate models** shows that **CatBoost** and **XGBoost** are the top performers for predicting costs. **CatBoost** achieved the lowest error with an **average RMSE of 1148.53** and a very high **average R^2 of 0.9479**, meaning this multivariate approach explains nearly **95% of the data variance**. While all models performed exceptionally well at predicting **Visa Fees** (with scores above 0.99), the multivariate ensemble methods like CatBoost significantly outperformed the **ElasticNet baseline**, especially in complex categories like **Rent** and **Insurance**. Overall, these results prove that analyzing multiple variables simultaneously with advanced boosting algorithms provides highly accurate and reliable predictions across all major expense categories.

	Avg_RMSE	Avg_R2	R2_Tuition_USD	R2_Rent_USD	R2_Insurance_USD	R2_Visa_Fee_USD
Model						
CatBoost (Multivariate)	1148.5326	0.9479	0.9260	0.9326	0.9421	0.9911
XGBoost (Multivariate)	1162.6586	0.9486	0.9239	0.9392	0.9377	0.9937
Random Forest (Multivariate)	1196.2124	0.9400	0.9200	0.9306	0.9212	0.9884
Gradient Boosting (Multivariate)	1197.6581	0.9455	0.9192	0.9386	0.9300	0.9941
ElasticNet (Baseline)	1244.5782	0.9248	0.9146	0.8689	0.9429	0.9729

Figure 21: Final Multivariate Model Selection using Key Predictors

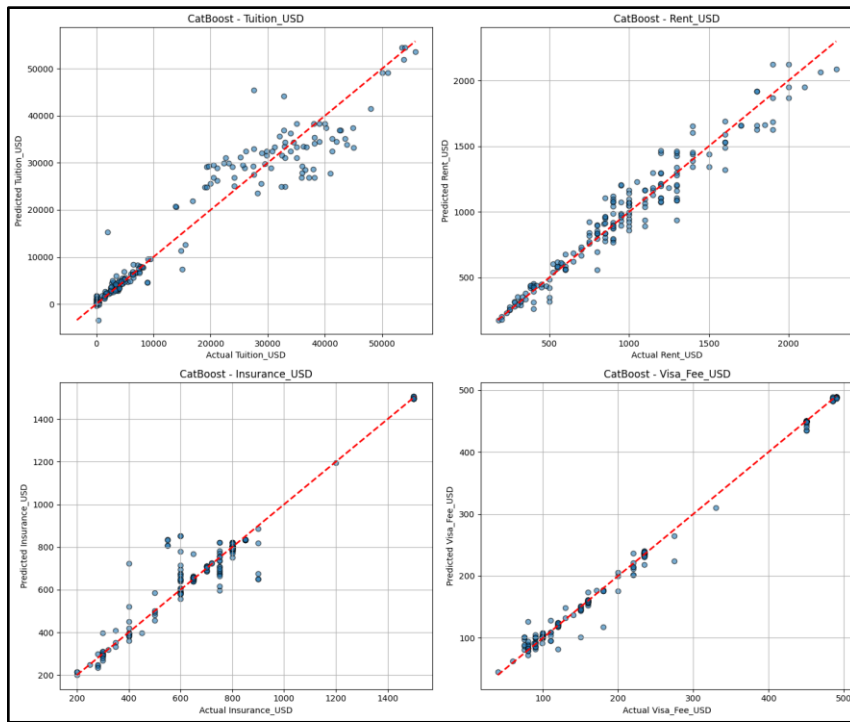


Figure 22: Scatter Plots of Predicted vs. Actual Values(In multivariate modeling)

The visual data confirms these high performance levels, specifically highlighting how accurately the **multivariate CatBoost model** maps predictions against actual costs. Across the four major expense categories **Tuition**, **Rent**, **Insurance**, and **Visa Fees** the scatter plots show data points tightly clustered along the 45-degree red dashed line, which represents a perfect match between predicted and actual values. This precision is particularly evident in the **Visa_Fee_USD** plot, where the model achieves near-

perfect alignment. This level of accuracy was reached through a rigorous selection process that prioritized retaining all data, as univariate analysis revealed that variables like the **Exchange Rate** contain a high percentage of outliers (26.66%) that are essential for realistic modeling. By choosing a multivariate CatBoost approach that includes these critical "real-world" extremes, the model maintains a superior **Test R-Squared of 0.9486**, ensuring reliable cost transparency for the entire international education journey.

12. Best model selection

Our model selection process followed a systematic path from initial data cleaning to the discovery of a high-performing multivariate solution. After finding that using the Living Cost Index as a predictor outperformed using it as a target, we tested the impact of outliers and found that keeping them was essential for maintaining a high R^2 score. By applying SHAP analysis to pinpoint the most influential features specifically Country, Living Cost Index, and Exchange Rate we were able to streamline our variables without losing accuracy. Ultimately, the **Multivariate CatBoost model** was chosen as the best performer, achieving an impressive R^2 of **0.9483** and the lowest error rate with a Test RMSE of **\$5226.86**.

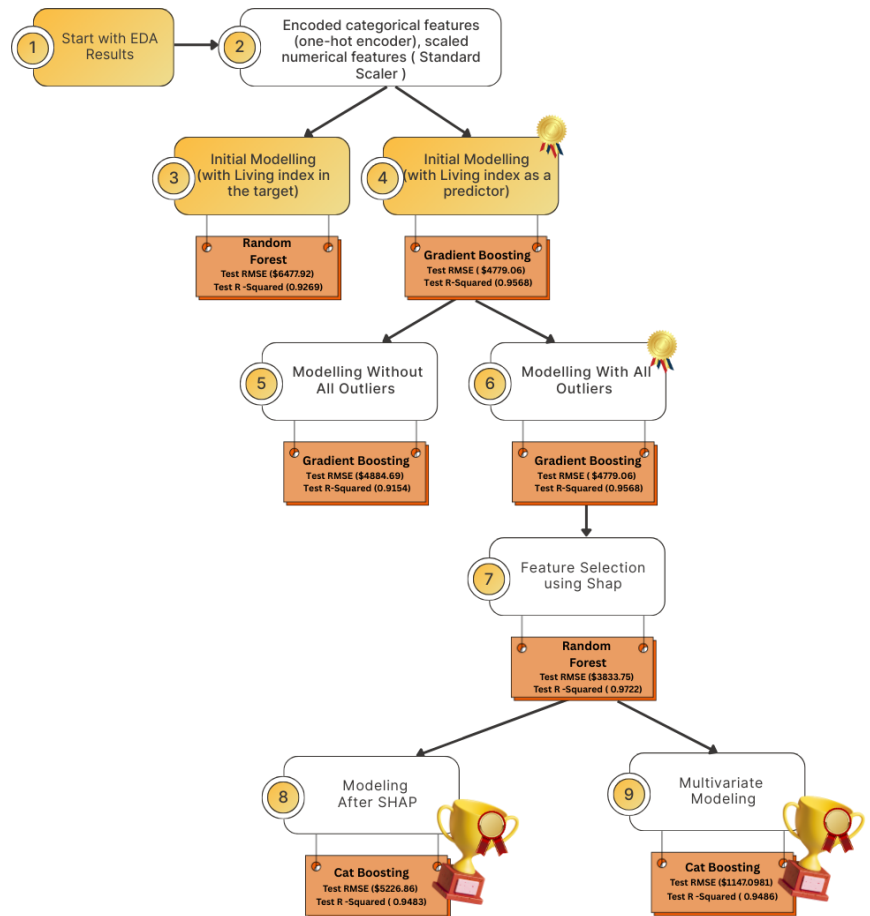


Figure 23: Best model selection (Full Procedure)

13. Final Product (EduCost Card Builder)

To turn our research into a practical tool, we developed the **EduCost Card Builder**, a specialized product designed for education agencies to streamline cost estimations. The system utilizes two distinct **CatBoost** models to provide a comprehensive financial breakdown: the total cost is generated by a univariate model, while individual expense categories are predicted using a multivariate model. This dual-model approach ensures that agencies can quickly set up accurate, data-driven profiles that reflect both the overall journey cost and the specific nuances of tuition, rent, and living expenses.

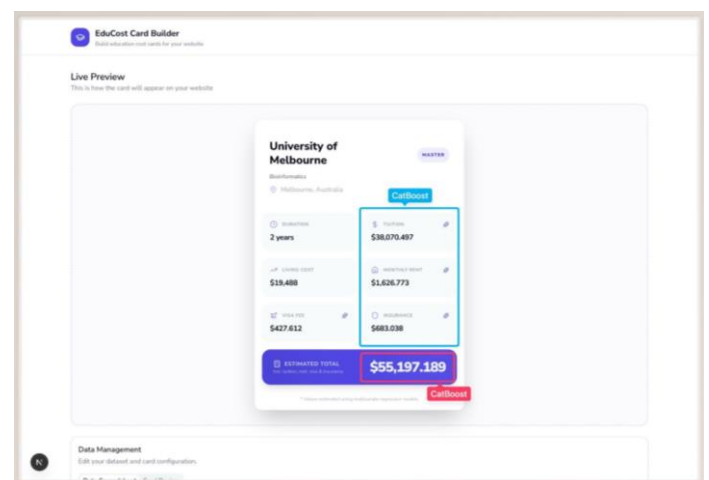


Figure 24: Final Product (EduCost Card Builder)

14. Issues encountered and proposed solutions

Table 2: Issues encountered and proposed solutions

Problem	Solution
The dataset provided the "Living Cost Index" variable (baseline: New York = 100) rather than actual living costs in currency units. This created a conceptual and analytical challenge during the project.	Initially, we attempted to convert this index into real costs using a reference baseline, but during the modeling process, we understood that the raw index works much better as a predictor
Non-Linear Relationships	Use Random Forest / XGBoost that handle non-linearity
outliers in several variables	In cost prediction, extreme values often represent real market segments, not data errors, and removing them would make the model incomplete and biased toward average-cost programs only
Balancing Project Work with other Academic Commitments	We maintained a shared Google Colab notebook where each team member contributed whenever they had free time between classes and assignments, allowing flexible asynchronous work. Everyone could see real-time progress, add code/visualizations, and leave comments for others.

15. Discussion and Conclusion

This project developed a machine learning model to predict the total cost of international education while addressing several practical challenges such as handling the Living Cost Index correctly, keeping real-world outliers, and reducing overfitting through iterative modeling and feature selection. The final models achieved high accuracy and were successfully translated into an interactive EDUCOST tool, showing the real-world usefulness of the work. However, there is strong potential for further improvement. Future work can expand the dataset by adding more countries, universities, and recent data to improve generalization. Including time-related economic factors such as inflation, currency fluctuations, and yearly tuition growth would allow the model to make future cost forecasts instead of only current estimates. Adding lifestyle-related variables such as accommodation type, spending habits, scholarships, and part-time job opportunities would help produce more personalized predictions for students. Model performance for very high-cost programs can be improved using larger datasets, advanced ensemble or stacking techniques, and deeper hyperparameter tuning. In addition, the tool can be developed into a complete decision-

support platform by integrating scholarship suggestions, budget planning dashboards, and risk analysis features. These improvements would increase the accuracy, usability, and real-world impact of the system.

16. Appendix

Link for the dataset:

<https://www.kaggle.com/datasets/adilshamim8/cost-of-international-education>

The python code used in this project:

<https://colab.research.google.com/drive/1aPPx0YRzOCfdLxBHXKu2Y-wk4unRn0-x?usp=sharing#scrollTo=cfrzGuCE5ffh>

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